Stock Analyzer & Price Predictor Using Machine Learning

1. Abstract

The stock market is a highly dynamic and complex system influenced by numerous economic, political, and psychological factors. Accurate stock price prediction is challenging yet valuable for investors and analysts. This project presents a **Stock Analyzer and Price Predictor** that leverages **machine learning techniques** to analyze historical stock market data and predict the **next-day closing price** of a selected stock.

The system fetches real-time and historical data from Yahoo Finance, performs technical analysis using moving averages and returns, trains a **Random Forest Regressor**, and visualizes actual versus predicted prices through an interactive web interface built with **Streamlit**. The project demonstrates how machine learning can assist in financial analysis while emphasizing its educational nature.

2. Introduction

Stock market analysis plays a crucial role in investment decision-making. Traditional analysis methods often rely on manual interpretation of charts and financial indicators, which can be time-consuming and subjective. With advancements in data science and machine learning, automated systems can now analyze large volumes of market data efficiently.

This project aims to:

- Provide real-time stock analysis
- Automate technical indicator computation
- Apply machine learning for predictive modeling
- Offer an interactive visualization platform

3. Objectives

The main objectives of this project are:

1. To collect live and historical stock market data
2. To perform technical analysis using statistical indicators
3. To train a machine learning regression model
4. To evaluate model performance using standard metrics
5. To predict the next trading day's stock price
6. To build a user-friendly web application for analysis

4. Technology Stack

Component	Technology
Programming Language	Python
Data Collection	yFinance
Data Processing	Pandas, NumPy
Visualization	Matplotlib
Machine Learning	Scikit-Learn
Web Framework	Streamlit

5. Dataset Description

- **Source:** Yahoo Finance
- **Data Type:** Time-series stock data
- **Attributes Used:** Open, High, Low, Close, Volume
- **Derived Features:**
  - 20-day Moving Average (MA20)
  - 50-day Moving Average (MA50)
  - Daily Returns

6. Methodology

Data Collection- Historical stock data is fetched using the yFinance library based on the user-provided ticker symbol.

Model Selection- *Random Forest Regressor* is chosen due to its:

- Ability to handle nonlinear relationships
- Robustness against noisy financial data
- Resistance to overfitting

7. Evaluation Metrics

The model is evaluated using:

- **RMSE (Root Mean Squared Error):**
  Measures the magnitude of prediction errors.
- **R^2 Score:**
  Indicates how well the model explains variance in stock prices.

8. Implementation Details

8.1 Web Application:

- Built using Streamlit
- Allows users to input stock ticker symbols
- Displays charts and prediction results interactively

8.2 Visualization

- Price trend graphs
- Actual vs Predicted comparison
- Technical indicator overlays

9. Results & Discussion:

The results obtained from the implementation of the Stock Analyzer and Price Predictor indicate that the Random Forest Regressor is capable of capturing general trends in stock price movements with reasonable accuracy. The comparison between actual and predicted prices shows that the model performs well during stable market conditions, producing predictions that closely follow real price movements. However, during periods of high volatility, prediction accuracy tends to decrease due to sudden market fluctuations and external factors not included in the dataset. Overall, the visualization of actual versus predicted values demonstrates the effectiveness of machine learning techniques in short-term stock price forecasting.

10. Limitations:

Despite achieving satisfactory results, the project has several limitations. The prediction model relies solely on historical price-based features and does not incorporate external factors such as news sentiment, macroeconomic indicators, or company financial fundamentals, which significantly influence stock prices. The system is designed only for next-day price prediction and does not support long-term forecasting. Additionally, market volatility and unexpected global events can negatively impact prediction accuracy, making the model unsuitable for real-world trading or investment decision-making.

11. Applications:

This project has multiple practical and educational applications. It can be used as a learning tool for students to understand the application of machine learning in financial markets and time-series data analysis. The system also serves as a demonstration platform for showcasing data preprocessing, feature engineering, model training, and evaluation techniques. Furthermore, it can be utilized for personal experimentation with stock data and as a foundation for more advanced financial analytics systems.

12. Future Enhancements:

Future enhancements of this project include the integration of additional technical indicators such as RSI, MACD, and Bollinger Bands to improve feature richness and predictive accuracy. The system can be extended to use deep learning models like LSTM or GRU, which are better suited for time-series forecasting and can capture long-term dependencies in stock price movements. Multi-day forecasting, portfolio performance analysis, real-time prediction updates, and cloud-based deployment using platforms such as AWS or GCP are also potential improvements that would enhance scalability, usability, and real-world relevance of the application.

13. Conclusion

This project demonstrates the application of machine learning in stock market analysis through an interactive and user-friendly platform. While the predictions are not intended for real-world trading, the system successfully showcases how data science techniques can extract insights from financial data. The project serves as a strong foundation for further exploration into financial analytics and predictive modeling.

14. References:

1. Yahoo Finance Documentation
2. Scikit-Learn Official Documentation
3. Streamlit Documentation
4. Introduction to Machine Learning – Coursera